

Quick Check

Lesson 8: Measuring Growth: Seedling Height, Stem Width and Growth Curves of Maize and Beans

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.2: Growth and Development in Plants | Lesson 8

Before measuring anything, a learner predicts that a maize seedling will grow the same number of centimetres every day for 21 days. Which statement about real seedling growth best challenges this prediction?

- A** Growth happens only in the first three days
- B** Growth keeps getting faster every day until the plant dies
- C** Growth is slow at first, then rapid, then slows down and levels off
- D** Growth is steady, so the graph is a straight line

A group wraps thread once around a bean stem. The thread measures 12 mm against a ruler. Using the method of dividing the circumference by 3, what is the approximate stem width (diameter)?

A 12 mm

B 9 mm

C 36 mm

D 4 mm

A group measures three maize seedlings from the class nursery. Their heights are 30 cm, 32 cm and 40 cm. What is the mean height?

A 34 cm

B 32 cm

C 102 cm

D 40 cm

Mean height data: maize was 8 cm on Day 6 and 48 cm on Day 15. Beans were 3 cm on Day 6 and 17 cm on Day 15. Between Day 6 and Day 15, how many more centimetres did the maize grow than the beans?

A 40 cm

B 14 cm

C 26 cm

D 31 cm

Mean maize height increased from 18 cm on Day 9 to 32 cm on Day 12. It then went from 58 cm on Day 18 to only 62 cm on Day 21. What does the flatter part of the curve between Day 18 and Day 21 show?

- A** The seed has not yet absorbed water
- B** The stationary phase, where growth slows as the plant matures or resources limit it
- C** The rapid phase, where growth is fastest
- D** The plant has died

Maize and bean seedlings were planted on the same day after the same rain. The two growth curves show that growth is not constant, and that the two crops grow at different rates. Why does this lead to the question 'Do we have hormones in plants?'

- A** Because growth speeds up and slows down at different times, suggesting something inside the plant may control its timing and speed
- B** Because seeds that failed to germinate have no growth curve
- C** Because the curves are straight lines
- D** Because plants make insulin like animals do

In a revised model of early growth, arrows are drawn showing food stores feeding the young seedling. Which pair correctly shows where maize and bean seedlings get these early food stores?

A Maize from the endosperm, beans from the cotyledons

B Both from hormones in the soil

C Both only from photosynthesis in the first days

D Maize from the cotyledons, beans from the endosperm

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).